

IOT BASED LEAF DISEASE DETECTION AND ANALYSIS USING IMAGE PROCESSING

A PROJECT REPORT

**Submitted in partial fulfilment of requirements for the
Award of the degree**

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATION ENGINEERING

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(Autonomous)**

**(Approved by AICTE, Permanently Affiliated to JNTUK, Kakinada,
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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
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CERTIFICATE



This is to certify that the project work entitled **IOT BASED LEAF DISEASE DETECTION AND ANALYSIS USING IMAGE PROCESSING** is the bonafide work carried out by **Ms. K. NANCY SUSHMA (16471A0488)**, **Ms. M. MADHAVI (16471A0499)**, **Mr. Sk. NAGOOR MASTHAN VALI (16471A04B5)**, **Mr. V. SAI PRAKASH (16471A0492)** and **Mr. M. V. POTHU RAJU (16471A0479)** in partial fulfilment of the requirements for the award of the degree Bachelor of Technology in Electronics and Communication Engineering from Jawaharlal Nehru Technological University Kakinada, Kakinada during the year 2019-2020 under my supervision and guidance.

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ABSTRACT

Agriculture plays an important role in every human life. It is an important sector of Indian economy which contributes about 17% of the total GDP. Also, agriculture provides employment to over 60% of the population directly or indirectly. But in contrast with the past, number of farmers have quit this field to other sectors due to some discomforts in farming. These diseases cost as social, ecological and economical loss to farmers. The main reasons for this involve high requirement of manpower, cost and time. Solving major issues like detecting effected plant and the level of occurrence requires more manpower and time that results in high cost. Hence, to overcome these aforementioned problems, IOT along with image processing is used for the disease detection and analysis. In understanding and analysing information from the images, the technique image processing has the utmost importance where the processes like modifying or interpreting of the existing pictures can be done. Disease in plants occur due to lack of identification in the initial stages. As the disease can attack and develop in any part of the plant in the whole field, it is difficult to test and recognize manually.

A perfect combination of Image processing with Internet of Things can produce new dimensions in the field of smart farming. This methodology aims for the approach of plant disease detection and analysis based on clustering techniques. This approach combines image processing and Internet of Things which runs on pre-processing and feature extraction techniques that helps to identify the plant effected area.

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LIST OF ABBREVIATIONS

ANN	Artificial Neural Networks
ASCII	American Standard Code for Information Interchange
IoT	Internet of Things
LAN	Local Area Network
LCD	Liquid Crystal Display
PCB	Printed Circuit Board
PWM	Pulse Width Modulation
RAM	Random Access Memory
SVM	Support Vector Machines

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CHAPTER – 1
INTRODUCTION

1.1 Introduction

Agriculture has played a key role in the development of human civilization. If there is decrease in agricultural products, total economy will get affected. As diseases are inevitable, detecting them plays major role. One can refer incident that occurred in 2007, Georgia (USA), it is estimated that approximately 539 USD was the loss incurred due to plant diseases as well as controlling them. The naked eye observation of farmers followed by chemical test is the main way of detection and classification of agricultural plant diseases. In developing countries, farming land can be much larger and farmers cannot observe each and every plant, every day. Farmers are unaware of non-native diseases. Consultation of experts for this might be time consuming & costly. Also, unnecessary use of pesticides might be dangerous for natural resources such as water, soil, air, food chain etc. as well as it is expected that there need to be less contamination of food products with pesticides.

Usually, for the continuous monitoring and detection of leaf diseases so much man power and time is essential. So, there is need for developing technique such as automatic plant disease detection and classification using leaf image processing techniques. This will prove useful technique for farmers and will alert them at the right time before spreading of the disease over large area. So, in the proposal an IoT based environment is developed for the continuous monitoring of the diseases of a farm and also to alert the relevant farmers. To predict the amount of disease affected for a leaf image processing technique are used.

1.1.1 Role of Agriculture in Indian Economy

Agricultural sector plays a strategic role in the process of economic development of a country. The agriculture sector is the backbone of an economy which provides the basic ingredients to mankind and now raw material for industrialisation.

It has already made a significant contribution to the economic prosperity of advanced countries and its role in the economic development of less developed countries is of vital importance.

“Increase in agricultural production and the rise in the per-capita income of the rural community, together with the industrialisation and urbanisation, lead to an increased demand in industrial production”-Dr. Bright Singh.

It is seen that increased agricultural output and productivity tend to contribute substantially to an overall economic development of the country, it will be rational and appropriate to place greater emphasis on further development of the agricultural sector.

Hence, to increase the effective crop yield it is necessary to monitor the crops continuously.

1.1.2 Introduction to IOT

The ‘Thing’ in IoT can be any device with any kind of built-in-sensors with the ability to collect and transfer data over a network without manual intervention. The embedded technology in the object helps them to interact with internal states and the external environment, which in turn helps in decisions making process.

IoT is a concept that connects all the devices to the internet and let them communicate with each other over the internet. IoT is a giant network of connected devices all of which gather and share data about how they are used and the environments in which they are operated.

IoT allows devices to be controlled remotely across the internet, thus it created opportunities to directly connect & integrate the physical world to the computer-based systems using sensors and internet. The interconnection of these multiple embedded devices will be resulting in automation in nearly all fields and also enabling advanced applications. This is resulting in improved accuracy, efficiency and economic benefit with reduced human intervention. It encompasses technologies such as smart grids, smart homes, smart irrigation, intelligent transportation and smart cities.

1.1.3 Introduction to Image Processing

Image Processing is a technique to enhance raw images received from cameras/sensors. Various techniques have been developed in Image Processing during the last four to five decades. Most of the techniques are developed for enhancing images obtained from unmanned spacecrafts, space probes and military reconnaissance flights. Image Processing systems are becoming popular due to easy availability of powerful personnel computers, large size memory devices, graphics, software etc.

The prevention and control of plant disease have always been widely discussed because plants are exposed to outer environment and are highly prone to diseases. Normally, the accurate and rapid diagnosis of disease plays an important role in controlling plant disease, since useful protection measures are often implemented after correct diagnosis. This system is based on image processing technology and uses MATLAB as the main processing tool.

The various Image Processing techniques are:

- Image representation
- Image pre-processing
- Image enhancement
- Image restoration
- Image analysis
- Image reconstruction
- Image data compression

Among these pre-processing and image analysis using k-means algorithm are used here.

1.2 Objective of the Project

Leaf disease is one of the most common reasons that results in low crop yield. Hence, a well-defined system is essential to ensure continuous monitoring and

early detection of leaf disease. Imaging leaves with more accuracy place a crucial role in diagnosis of leaf disease. The main aim of our system is to provide a simple and easiest method to identify meaningful and accurate information of leaf image and detecting the disease in order to take immediate action.

1.3 Literature Survey

All the existing methods for detection and analysis of leaf disease involves any one of the below different methods.

- Manual Detection
- Threshold based Segmentation
- Leaf Disease Detection using Deep Learning
- Plant Disease Recognition based on Image Processing
- Leaf Disease Analysis using k-means Segmentation
- Detection of Plant Disease using Neural-networks-based Classification

Manual detection involves in examining the leaves by naked human eye or by using a microscope. Though it is highly difficult that consumes more time and manpower, it is used in the olden days, when technology is not that much developed.

Thresholding technique is based on a threshold value to turn a gray-scale image into a binary image. In this technique image is segmented by comparing pixel values with the predefined threshold limit. The limitation is the problem in differentiating the pixels from some healthy pixels due to the fact that some of the pixels in the leaf appeared to have a similar color to the diseased area.

Clustering techniques involves number of approaches. One of them is the *K-means* clustering that aims to partition of n observations into k clusters in which each observation belongs to the cluster with the nearest mean, serving as a prototype of the cluster.

Deep neural networks have recently been successfully applied in many diverse domains as examples of end to end learning. Neural networks provide a

mapping between an input—such as an image of a diseased plant—to an output—such as a crop disease pair. The nodes in a neural network are mathematical functions that take numerical inputs from the incoming edges, and provide a numerical output as an outgoing edge. Deep neural networks are simply mapping the input layer to the output layer over a series of stacked layers of nodes. The challenge is to create a deep network in such a way that both the structure of the network as well as the functions (nodes) and edge weights correctly map the input to the output. Deep neural networks are trained by tuning the network parameters in such a way that the mapping improves during the training process. This process is computationally challenging and has in recent times been improved dramatically by a number of both conceptual and engineering breakthroughs. Manual segmentation gives the best and most reliable results when identifying structures for a particular clinical task. Usually, a medical practitioner with knowledge of anatomy utilizes a mouse-based software to outline or fill regions of target structures on each image slice in an image stack, i.e. a volume. The segmenter needs to mentally reconstruct a structure in 3D because only 2D image slices can be viewed. Segmentation of each slice requires that the segmenter view adjacent images in the stack in order to correctly reconstruct an object in 3D. The task is very tedious and time consuming for the segmenter, and thus does not serve the need of daily clinical use as well.

Neural networks, with their outstanding ability to derive meaning from complex or imperfect data, can be applied for extracting patterns and detecting trends that are too difficult to notice by humans or computer techniques. Other advantages of ANNs are adaptive learning, self-organization, real time operations, and so forth. This is a new approach in detecting plant diseases using the deep convolutional neural network trained and fine-tuned to fit accurately to the database of a plant's leaves that was gathered independently for diverse plant diseases. The advance and novelty of the developed model lie in its simplicity; healthy leaves and background images are in line with other classes, enabling the model to distinguish between diseased leaves and healthy ones or from the environment by using deep CNN.

1.4 Organization of the project

Introduction to leaf disease detection and analysis which gives the main objective of the project and literature survey on some existing methods is discussed in chapter 1, Existing methods which include several techniques to detect the leaf diseases are described in chapter 2. The proposed method which is chapter 3 describes pre-processing technique and posterior analysis. The final results obtained from the proposal are discussed in chapter 4. The last chapter i.e. chapter 5 have the conclusions and future scope for the project.

CHAPTER - 2

EXISTING METHODS

2.1 Introduction

In leaf disease analysis, experts integrate their knowledge under microscopic analysis to obtain the nature and pathological characteristics of leaf and to decide on treatment options. However, in microscopic or with naked eye testing requires a great number of persons to test for every plant. Physical detecting and segmenting the effected leaves is monotonous. Therefore, there is a need for computer aided device that helps in effected leaf detection and segmentation from the images to overcome the problems involved in the manual segmentation. Number of methods has been proposed in recent years to seal this break, but still there is no generally customary automated technique to be used in clinical floor due to accuracy and robustness issues. Artificial Intelligence methods such as Digital Image Processing when cooperative with others like IoT, machine knowledge, fuzzy logic and pattern recognition are so valuable in Image processing techniques.

2.2 Existing Methods for Disease Detecton

In past years, various traditional machine learning algorithms have been used to disease diagnosis and analysis. Recently, several researchers have studied plant leaf disease detection and classification with neural network algorithms and image processing. Also, the research studies showed that convolutional neural network have been used majorly for plant disease recognition and showed good results.

Image segmentation normalization are the primary and most critical tasks of image analysis. The purpose of segmentation is that of extracting information from an image by means of image segmentation and normalization deals with adjusting the range of pixel intensity. Normalization is also known as contrast stretching and histogram stretching. The mechanization of medical image segmentation has established wide application in diverse areas such as verdict for patients, treatment management planning, and computer-integrated surgery.

Nowadays, leaf segmentation techniques can be composed into different classes based on different principles. Some of them are:

- Manual Detection
- Threshold based Segmentation
- Leaf Disease Detection using Deep Learning
- Plant Disease Recognition based on Image Processing
- Leaf Disease Analysis using k-means Segmentation
- Detection of Plant Disease using Neural-networks-based Classification

2.2.1 Manual Detection

Manual recognition is a process where leaves are diagnosed based on the experience by seeing with one's naked eye. But some problems occur: on the one hand, it can be mistakenly diagnosed by farmers because they usually judge the symptom by their experiences; on the other hand, the disease treatment may be dallied over because the technician or expert can't go to locale to diagnose in good time. Also, experts started using microscopes to diagnosing the leaves. But detection under microscope takes much time. Relative to the person's vision, computer image processing technique take on some characteristics such as speediness, huge information and distinguish small diversity which can't be distinguished by person's eyes, so image processing technique can help farmers to judge the reasons and severity of crop diseases, and it takes on important theoretical and practical significance for improving the automatic management of crop. The research on crop disease recognition with image processing technology started latterly, which is mainly concentrated in the 80s to 90s years of 20th century and the beginning of 21st century.

2.2.2 Deep Learning

In statistical machine learning, a major issue is the selection of an appropriate feature space where input instances have desired properties for solving a particular problem. For example, in the context of supervised learning for binary classification, it is often required that the two classes are separable by a hyperplane. In the case where this property is not directly satisfied in the input space, one is given the possibility to map instances into an intermediate feature space where the classes are linearly separable. This intermediate space can either

be specified explicitly by hand-coded features, be defined implicitly with a so-called kernel function, or be automatically learned. In both of the first cases, it is the user's responsibility to design the feature space. This can incur a huge cost in terms of computational time or expert knowledge, especially with highly dimensional input spaces, such as when dealing with images. As for the third alternative, automatically learning the features with deep architectures, i.e. architectures composed of multiple layers of nonlinear processing, can be considered as a relevant choice. Indeed, some highly nonlinear functions can be represented much more compactly in terms of number of parameters with deep architectures than with shallow ones (e.g. SVM). For example, it has been proven that the parity function for n -bit inputs can be coded by a feed-forward neural network with $O(\log n)$ hidden layers and $O(n)$ neurons, while a feed-forward neural network with only one hidden layer needs an exponential number of the same neurons to perform the same task. Moreover, in the case of highly varying functions, learning algorithms entirely based on local generalization are severely impacted by the curse of dimensionality. Deep architectures address this issue with the use of distributed representations and as such may constitute a tractable alternative.

Unfortunately, training deep architectures is a difficult task and classical methods that have proved effective when applied to shallow architectures are not as efficient when adapted to deep architectures. Adding layers does not necessarily lead to better solutions. For example, the more the number of layers in a neural network, the lesser the impact of the back-propagation on the first layers. The gradient descent then tends to get stuck in local minima or plateaus, which is why practitioners have often preferred to limit neural networks to one or two hidden layers.

Plant Disease Recognition based on Image Processing

The prevention and control of plant disease have always been widely discussed because plants are exposed to outer environment and are highly prone to diseases. Normally, the accurate and rapid diagnosis of disease plays an important role in controlling plant disease, since useful protection measures are

often implemented after correct diagnosis. This system is based on image processing technology and uses MATLAB as the main processing tool. Besides, digital image processing, mathematical statistics, plant pathology, and other relative fields are also considered. Comparing to the traditional image recognition, there are plenty of innovations in image segmentation and system construction. To strengthen the division of the lesion, users have different creative interactive options to meet their own needs. Meanwhile, line AR regression model can be used in various types of plant disease.



Fig. 2.1: Flow chart for Disease Detection

Image Analysis Can Be Applied For The Following Purposes:

1. To detect diseased leaf, stem, fruit.
2. To quantify affected area by disease.
3. To find the boundaries of the affected area.
4. To determine the color of the affected area.
5. To determine size & shape of leaf.
6. To identify the Object correctly.

Etc.

2.2.3 Histogram Segmentation Method

Segmentation refers to the process of partitioning a digital image into the multiple segments (set of pixels as known as super pixels) the goal of segmentation is to simplify and or change the representation of an image into something that is more meaningful and easier to analyze. Development of an accurate image segmentation algorithm can be most demanding part of a

computer vision system there is not a panacean method that can be work with several different types of images in the segmentation approach is usually designed for solving a specific problem. In this work, histogram thresholding is proposed in order to help the segmentation step in what was found to be robust way regardless of the segmentation approach used semi atomic algorithm for histogram thresholding are discussed. Examples using different histogram thresholding Methods are shown. This segmentation method can automatically determine the threshold. It can greatly reduce the user's task burden and optimize the image segmentation process. The processing details are as follows; the flow diagram of the improved histogram segmentation is shown in fig. 2.2.

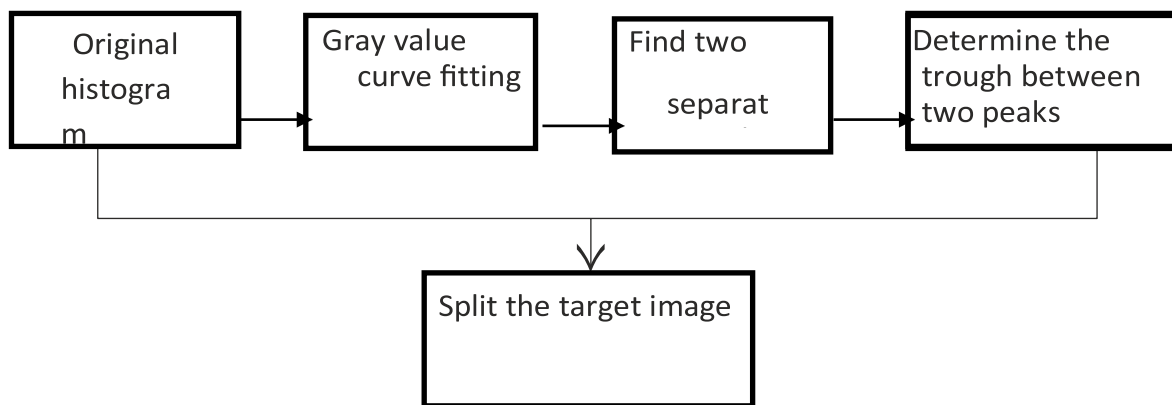


Fig. 2.2: The process of the histogram segmentation.

2.2.4 k-means Segmentation

Data clustering is the process of grouping data elements based on some aspects of relationship between the elements in the group. Clustering has many applications such as data firmness, data mining, pattern recognition, machine learning. There are many different clustering methods. K-means method of clustering select of primary seed for dividing a group of clusters that affects the result.

Clustering can be considered the most important unsupervised learning problem; so, as every other problem of this kind, it deals with finding a structure in a collection of unlabeled data. A loose definition of clustering could be “the process of organizing objects into groups whose members are similar in some way. A cluster is therefore a collection of objects which are “similar” between

them and are “dissimilar” to the objects belonging to other clusters. We can show this with a simple graphical example:

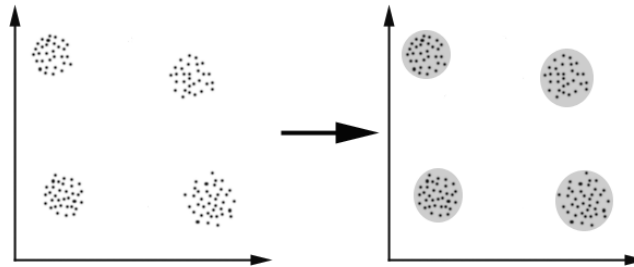


Fig. 2.3: Example of Clustering

In this case we easily identify the four clusters into which the data can be divided; the similarity criterion is distance: two or more objects belong to the same cluster if they are “close” according to a given distance (in this case geometrical distance). This is called distance-based clustering. Another kind of clustering is conceptual clustering two or more objects belong to the same cluster if this one defines a concept common to all that objects. In other words, objects are grouped according to their fit to descriptive concepts, not according to simple similarity measure.

2.2.5 Neural-networks

In this changing environment, appropriate and timely disease identification including early prevention has never been more important. There are several ways to detect plant pathologies. Some diseases do not have any visible symptoms, or the effect becomes noticeable too late to act, and in those situations, a sophisticated analysis is obligatory. However, most diseases generate some kind of manifestation in the visible spectrum, so the naked eye examination of a trained professional is the prime technique adopted in practice for plant disease detection. In order to achieve accurate plant disease diagnostics a plant pathologist should possess good observation skills so that one can identify characteristic symptoms. Variations in symptoms indicated by diseased plants

may lead to an improper diagnosis since amateur gardeners and hobbyists could have more difficulties determining it than a professional plant pathologist. An automated system designed to help identify plant diseases by the plant's appearance and visual symptoms could be of great help to amateurs in the gardening process and also trained professionals as a verification system in disease diagnostics. Advances in computer vision present an opportunity to expand and enhance the practice of precise plant protection and extend the market of computer vision applications in the field of precision agriculture. Exploiting common digital image processing techniques such as color analysis and thresholding were used with the aim of detection and classification of plant diseases. Various different approaches are currently used for detecting plant diseases and most common are artificial neural networks (ANNs) and Support Vector Machines (SVMs). They are combined with different methods of image preprocessing in favor of better feature extraction. In machine learning and cognitive science, ANN is an information-processing paradigm that was inspired by the way biological nervous systems, such as the brain, process information. The brain is composed of a large number of highly interconnected neurons working together to solve specific problems. An artificial neuron is a processing element with many inputs and one output. Although artificial neurons can have many outputs, only those with exactly one output will be considered. Their inputs can also take on any value between 0 and 1. Also, the neuron has weights for each input and an overall bias. The weights are real numbers expressing importance of the respective inputs to the output. The bias is used for controlling how easy the neuron is getting to output 1. For a neuron with really big bias it is easy to output 1, but when the bias is very negative then it is difficult to output 1. The output of the neuron is not 0 or 1. Instead, it is $\alpha \cdot (w \cdot x + b)$, where α is called the transfer function. There are different types of transfer function: step, linear, sigmoid, and so forth. The smoothness of α means that small changes Δw_j in the weights and Δb in the bias will produce small change Δ output in the output from the neuron. Small output change is approximated by

$$\Delta \text{output} \approx \sum \partial \text{output} / \partial w_j * \Delta w_j + \sum \partial \text{output} / \partial b * \Delta b$$

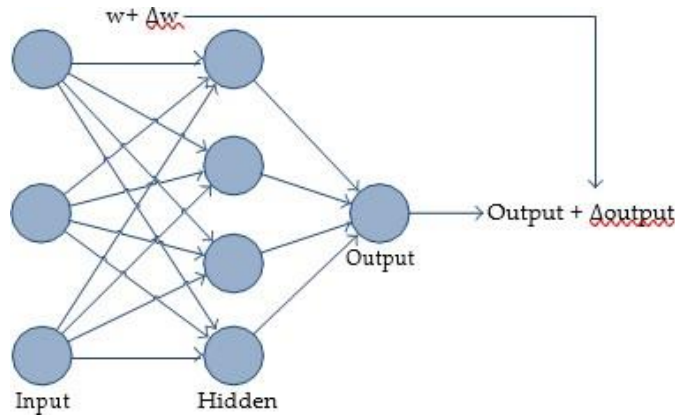


Fig. 2.4: Simple model of ANN

Basically, the small change in weight or bias causes the small corresponding change in the network output. Neural networks, with their outstanding ability to derive meaning from complex or imperfect data, can be applied for extracting patterns and detecting trends that are too difficult to notice by humans or computer techniques. Other advantages of ANNs are adaptive learning, self-organization, real time operations, and so forth. There are two main categories of ANNs when speaking about architecture: feed-forward ANNs where the output of any layer is unlikely to influence that same layer and feedback ANNs where signals travel in both directions by involving loops in the network.

The method described in this paper is a new approach in detecting plant diseases using the deep convolutional neural network trained and fine-tuned to fit accurately to the database of a plant's leaves that was gathered independently for diverse plant diseases. The advance and novelty of the developed model lie in its simplicity; healthy leaves and background images are in line with other classes, enabling the model to distinguish between diseased leaves and healthy ones or from the environment by using deepCNN.

CHAPTER - 3

PROPOSED METHOD & IMPLEMENTATION

3.1 Introduction to Proposed System

In past years, various traditional machine learning algorithms have been used to disease diagnosis and analysis. Recently, several researchers have studied plant leaf disease detection and classification with neural network algorithms and image processing. Also, the research studies showed that convolutional neural network have been used majorly for plant disease recognition and showed good results. The proposed approach combines the techniques of Internet of Things and Image processing concepts to get expected output.

3.2 Block Diagram

The proposed schematic of the IOT based leaf disease detection system is shown in fig.3.1. The block diagram represents the main blocks used in the project namely power supply, Camera, Raspberry Pi with a display unit.

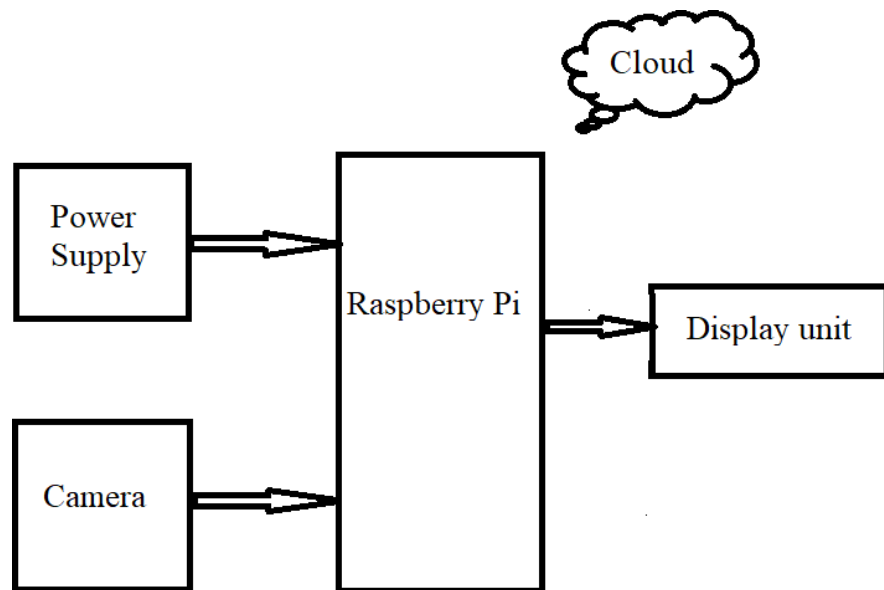


Fig. 3.1: Block diagram

- Raspberry pi
- Camera
- Display unit
- Power Supply

3.2.1 Raspberry Pi

RASPBERRY PI 3 is a development board in PI series. It can be considered as a single board computer that works on LINUX operating system. The board not only has tons of features it also has terrific processing speed making it suitable for advanced applications. PI board is specifically designed for hobbyist and engineers who are interested in LINUX systems and IOT (Internet of Things).



Fig. 3.2:Raspberry Pi

RASPBERRY PI platform is most used after ARDUINO. Although overall applications of PI are less it is most preferred when developing advanced applications. Also, the RASPBERRY PI is an open source platform where one can get a lot of related information so you can customize the system depending on the need.

Here are few examples where RASPBERRY PI 3 is chosen over other microcontrollers and development boards:

1. Where the system processing is huge. Most ARDUINO boards all have clock

speed of less than 100MHz, so they can perform functions limited to their capabilities. They cannot process high end programs for applications like Weather Station, Cloud server, gaming console etc. With 1.2GHz clock speed and 1 GB RAM RASPBERRY PI can perform all those advanced functions.

2. Where wireless connectivity is needed. RASPBERRY PI 3 has wireless LAN and Bluetooth facility by which you can setup WIFI HOTSPOT for internet connectivity. For Internet of Things this feature is best suited.

3. RASPBERRY PI also has dedicated camera port so one can connect camera without any hassle to the PI board.

4. RASPBERRY PI also has PWM outputs for application use.

There are many other features like HD steaming which further promote the use of RASPBERRY PI.

Raspberry Pi is an **ARM** based credit card sized **SBC** (Single Board Computer) created by Raspberry Pi Foundation. Raspberry Pi runs Debian based **GNU/Linux** operating system Raspbian and ports of many other OSes exist for this SBC.

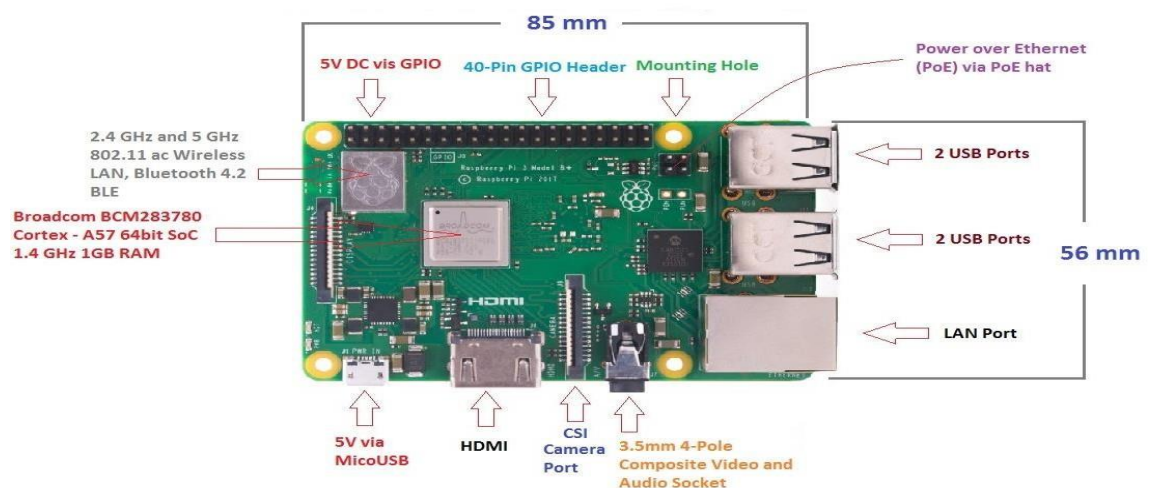


Fig. 3.3: Raspberry Pi 3 Model B pin description

Table 3.1: Pin Description of Pi 3 model B

Board	Raspberry Pi 3
Processor	Broadcom BCM 3287
CPU core	Quadcore ARM Cortex-A53, 64Bit
Clock Speed	1.2GHz (Roughly 50% faster than Pi2)
RAM	1 GB
GPU	400 MHz Video Core IV
Network Connectivity	1 x 10 / 100 Ethernet (RJ45 Port)
Wireless Connectivity	802.11n wireless LAN (Wi- Fi) and Bluetooth 4.1
USB Ports	4 x USB 2.0
GPIOs	2 x 20 Pin Header
Camera Interface	15-pin MIPI
Display Interface	DSI 15 Pin / HDMI Out / Composite RCA
Power supply(Current Capacity)	2.5 A

Raspberry Pi 3 B that was introduced in 2016 came with CPU, GPU, USB ports and I/O pins. **Raspberry Pi 3 B+**. It is a small board computer, introduced by Raspberry Pi foundation in 14th March 2018 and is the most recent version of the Pi boards. Both versions Raspberry Pi 3 B and Raspberry Pi 3 B+ are almost same in terms of functionality and technical specifications; however, there are some exceptions in the B+ model as it comes with USB boot, network boot, and Power over Ethernet option that are not present in the B model.

Improvements in B+ than B:

- The model B+ stays ahead in terms of processing speed and comes with an improved wireless capability.
- The dual-band WiFi 802.11ac runs at 2.4GHz and 5GHz and provides a better range in wireless challenging environments and Bluetooth 4.2 is available with BLE support.
- The top side is painted with metal shielding, instead of plastic in the earlier models, that acts as a heat sink and drains the excessive amount of heat if the board is subjected to the high temperature or pressure.
- This B+ model is three times faster than Pi 2 and 3 which is a major development in terms of speed, capable of executing different functions at a decent pace.
- The ethernet port comes with 300 Mbit/s which is much faster than earlier version with 100 Mbit/s speed. It is known as gigabit ethernet based on USB 2.0 interface.
- Four pin header is added on the board that resides near 40 pin header. This allows the Power over Ethernet (PoE) i.e. provides the necessary electrical current to the device using data cables instead of power cords. It is very useful and reduces the number of cables required for the installation of a device in the relevant project.
- PoE works only in the presence of PoE hat.

Raspberry Pi technical specifications:

- CPU is 64 bit with 1GB RAM (random access memory)
- Contains Broadcom BCM2837B0 chipset
- Comes with 1.4GHz Quad-Core ARM Cortex-A53, 4 cores
- Consists of 40 pin header (26 GPIOs)
- Stereo audio and composite video is supported by 3.5mm jack connector
- 4 USB 2.0 ports
- Gigabit Ethernet
- PoE (power over Ethernet) is a major feature incorporated in this device that lacks in B model
- 2-pin reset header
- Micro SD socket, used to enhance the memory capacity of the board
- MicroUSB power connector, used for transferring power to the device
- HDMI
- CSI camera interface
- Comes with WiFi and Bluetooth facility that were not present in previous Raspberry Pi 1 and 2 versions
- DSI connector for official screen

3.2.2 Camera

A camera is an optical instrument used to record images. At their most basic cameras are sealed boxes (the camera body) with small holes (the aperture) that let light in, to capture an image on a light-sensitive surface (usually photographic film or a digital sensor). In this project, the camera is used to capture the leaf images.

3.2.2.1 Interfacing Camera to Pi

Current models of Raspberry Pi have a port for connecting the Camera Module. The camera board attached to the Raspberry Pi via a 15-way ribbon cable. There are only two connections to make: the ribbon cable needs to be attached to the camera PCB, and to the Raspberry Pi itself. You need to get the cable the right way around, or the camera will not work. On the camera PCB, the blue backing on the cable should face away from the PCB, and the Raspberry Pi it should face towards the Ethernet connection (or where the Ethernet connector would be if you're using a model A).

There are two versions of the Camera Module:

- The standard version, which is designed to take pictures in normal light
- The NoIR version, which doesn't have an infrared filter, so you can use it together with an infrared light source to take pictures in the dark.

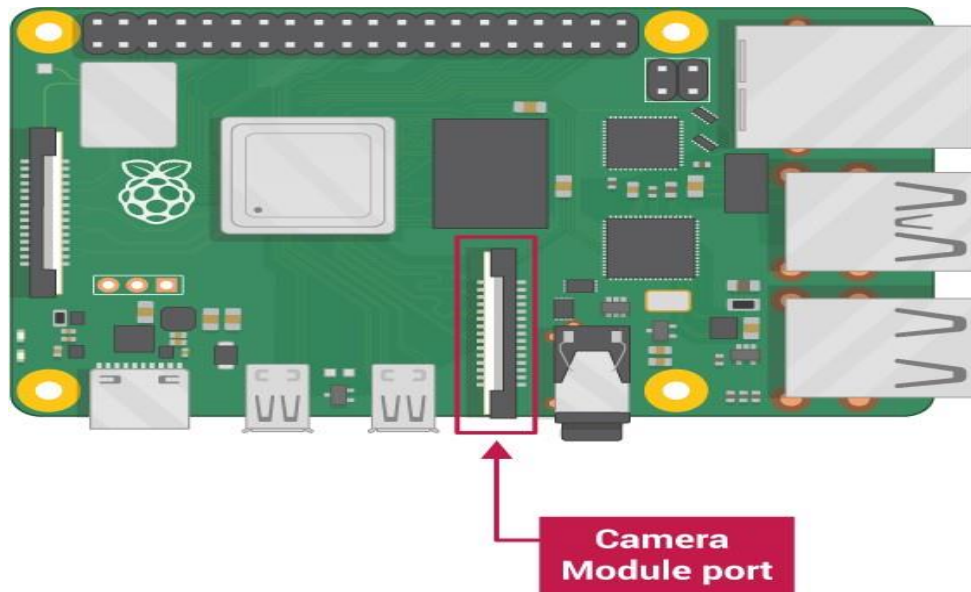


Fig. 3.4: Raspberry Pi with Camera module

Connect the Camera Module

Ensure your Raspberry Pi is turned off. Locate the Camera Module port

1. Gently pull up on the edges of the port's plastic clip.
2. Insert the Camera Module ribbon cable; make sure the cable is the right way around.
3. Push the plastic clip back into place.



Fig. 3.5: Interfacing camera to Pi board

- Start up the Raspberry Pi.
- Go to the main menu and open the Raspberry Pi Configuration tool.

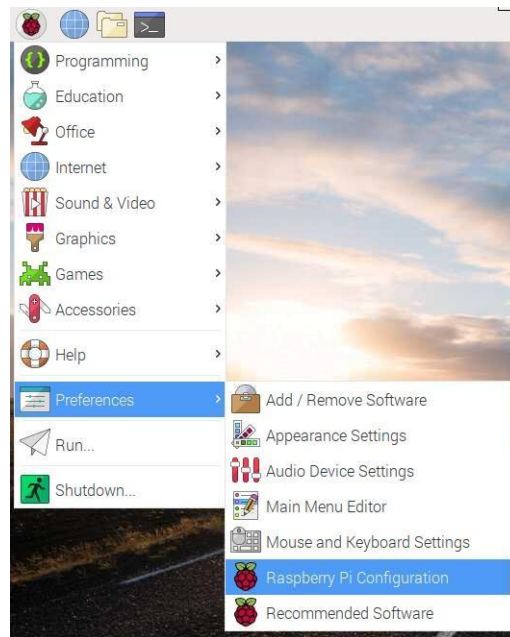


Fig. 3.6: Raspberry Pi Configuration tool

- Select the Interfaces tab and ensure that the camera is enabled:

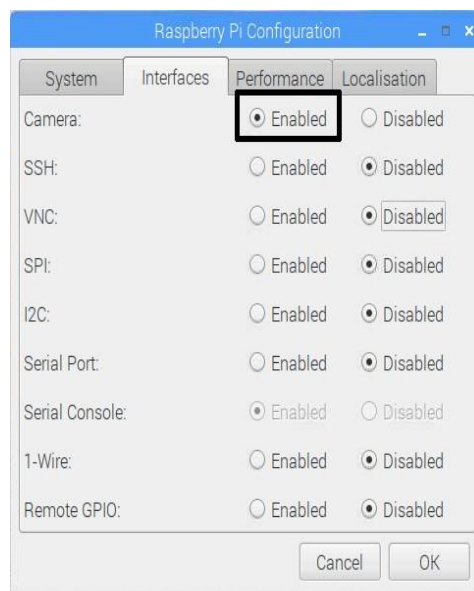


Fig. 3.7: Setting Interfaces

- Reboot the Raspberry Pi.

Now that the Camera Module is connected and the software is enabled, try out the command line tools `raspistill` and `raspivid`.

- Open a terminal window by clicking the black monitor icon in the taskbar:



- Type in the following command to take a still picture and save it to the Desktop:

```
raspistill -o Desktop/image.jpg
```

When the command runs, you can see the camera preview open for five seconds before a still picture is taken.

- Look for the picture file icon on the Desktop, and double-click the file icon to open the picture.

By adding different options, you can set the size and look of the image the `raspistill` command takes.

For example, add `-h` and `-w` to change the height and width of the image:

```
raspistill -o Desktop/image-small.jpg -w 640 -h 480
```

- Now record a video with the Camera Module by using the following `raspivid` command:

```
raspivid -o Desktop/video.h264
```

3.2.3 Display Unit

Display Unit is a device used to display images or information. Based on the application, different devices like LCDs or monitors can be used as a display device. Regardless of the device used a display unit is meant for displaying the output data. Here, monitor is used as display unit that displays leaf images and information regarding leaf disease.

A **monitor** is an electronic visual computer display that includes a screen, circuitry and the case in which that circuitry is enclosed. Older computer **monitors** made use of cathode ray tubes (CRT), which made them large, heavy and inefficient. A **monitor** is also known as a screen or a visual display unit (VDU).

A computer monitor is a **display** adapter that displays information processed by the computer's video card. When a video card or graphics card converts binary information from 1s and 0s into images, these images are displayed onto the directly connected monitor.



Fig. 3.8: Display Device

The monitor connected at the output side of the circuitry so that after the entire processing the output can be displayed on the monitor screen using HDMI

cable. The serial data from the Pi is converted into the parallel data while transmitting thorough the cable as, the monitor satisfies parallel data communication.

3.2.4 Power Supply

A power supply is a component that supplies power to at least one electric load. Typically, it converts one type of electrical power to another, but it may also convert a different form of energy – such as solar, mechanical, or chemical - into electrical energy. A power supply provides components with electric power.

A power supply is an electrical device that supplies electric power to an electrical load. The primary function of a power supply is to convert electric current from a source to the correct voltage, current, and frequency to power the load.

Most power supplies are made up of four basic sections: a TRANSFORMER, a RECTIFIER, a FILTER, and a REGULATOR. The schematic block diagram of a basic power supply is as shown in fig. 3.9. As shown, the first section in the power supply is the TRANSFORMER.

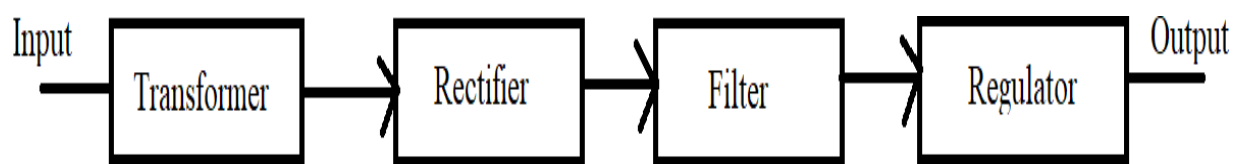


Fig. 3.9: Power Supply block diagram

The Raspberry Pi 3 is a nice small hobby computer along with built-in Wi-Fi and Bluetooth. All the extra components can make the Pi 3 more power hungry

than its predecessors. The original unit needed a 700mA supply. The requirements have slowly been moving up with each new release with the Pi 3 recommended to have a 2.5 Amp 5 Volt supply.

3.3 Image Acquisition using IoT

Image acquisition, it is defined as the action of retrieving an image from some source, usually a hardware-based source for processing. It is the first step in the workflow sequence because, without an image, no processing is possible. In this conceptual approach, this stage indicates the input. The images are retrieved by using camera. The leaf image is extracted and gets saved in the database for further process.

In this process, the leaf images captured by the camera are collected and saved in the memory which is called as image acquisition. Also, after the entire process is completed, the percentage of effected area in the leaf is uploaded to the server. Every input disease effected image is compared with the images in the cloud, so that it can detect the type of the disease.

3.4 Image Analysis:

Image Analysis is where the analysis of the effected leaf is done using image processing. This includes pre-processing and posterior analysis. Pre-processing of leaf includes normalization and posterior analysis is done by using K-means algorithm and feature selection for a leaf image.

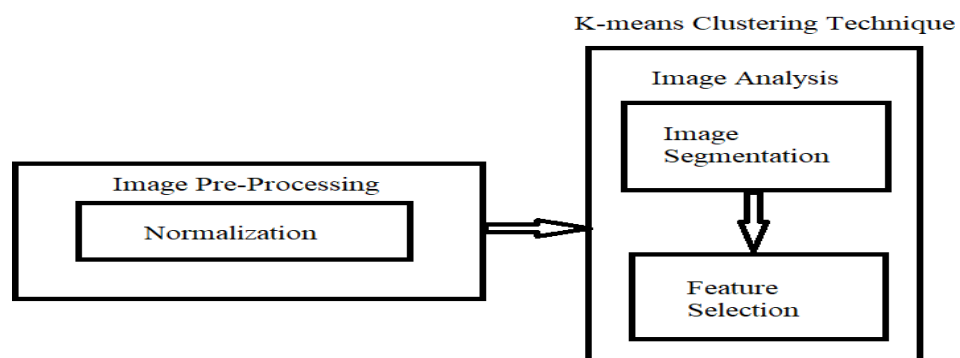


Fig. 3.10: Process of Image Analysis

3.4.1 Pre-Processing

Pre-processing is a common method for operations with images at the lowest level of abstraction both input and output are intensity images. The aim of pre-processing is an improvement of the image data that suppresses unwanted distortions or enhances some image features important for further processing. In this stage the noise removal and data normalization are used as pre-processing model in which all features are normalized from vector to unit space.

3.4.1.1 Normalization

Normalization is a process that changes the range of pixel intensity values. Applications include photographs with poor contrast due to glare, for example. Normalization is sometimes called contrast stretching or histogram stretching. At this stage the image pixels are normalized by having certain value as its threshold, which separates healthy and diseased region of the plant leaf by using the color of the leaf.

3.4.2 Posterior Analysis

Posterior analysis includes applying some clustering algorithms for the leaf pixels with some feature extractions. This is done once the pre-processing of the leaf images is done. Here, K-means algorithm is used. In this step, segmentation of images is done to find the region of interest i.e., the region of leaf effected.

3.4.2.1 k-means Clustering Technique

The k-means clustering algorithm attempts to split a given anonymous data set (a set containing no information as to class identity) into a fixed number (k) of clusters. Initially k number of so-called centroids are chosen. Each centroid is thereafter set to the arithmetic mean of the cluster it defines. K-means clustering is one of the simplest and popular unsupervised machine learning algorithms. Typically, unsupervised algorithms make inferences from datasets using only input vectors without referring to known, or labelled, outcomes. The objective of K-means is simple, group similar data points together and discover underlying

patterns. To achieve this objective, K-means looks for a fixed number (k) of clusters in a dataset

A cluster refers to a collection of data points aggregated together because of certain similarities. K is a target number, which refers to the number of centroids you need in the dataset. A centroid is the imaginary or real location representing the centre of the cluster. Every data point is allocated to each of the clusters through reducing the in-cluster sum of squares. In other words, the K-means algorithm identifies k number of centroids, and then allocates every data point to the nearest cluster, while keeping the centroids as small as possible. The '*means*' in the K-means refers to averaging of the data; that is, finding the centroid. The K-means algorithm in data mining starts with a first group of randomly selected centroids, which are used as the beginning points for every cluster, and then performs iterative (repetitive) calculations to optimize the positions of the centroids.

It halts creating and optimizing clusters when either:

- The centroids have stabilized, there is no change in their values because the clustering has been successful.
- The defined number of iterations has been achieved.

3.4.2.2 Feature Selection

Feature selection is a posterior process used for image analysis where the selected or desired features alone are considered from a set of features of leaf. Clearly, feature selection is a process where selected attributes of an examined leaf are considered. There are certain features for leaf like shape, colour, width, area etc., Here, we are considering the area of the leaf.

For example, consider the pixel intensities of an image varies in between 0 and 255. This image is first normalised, where it maintains a value as its threshold limit, consider 112 is the threshold limit for the considered leaf image. Then all the pixel values above this threshold intensity are taken as 1's and the remaining

are taken as 0's. After this there will be a clear separation between the effected and the non-effected leaf positions.

K-means algorithm is used for the further processing, which helps in grouping the same range of pixel intensities. Feature selection involves while the mean among the pixels is calculated. The total percentage affected area of the leaf is calculated by using the '.Area' command, it gives the percentage of effected area in each cluster.

A1=diseasedata.Area;

Later, the total effected percentage area of the leaf is calculated from the individual area percentages and will be displayed on the display unit.

3.5 Circuit and Set Up:

The Complete circuit connections and the setup is shown in below fig.3.11. For capturing leaf images, a camera is used and to process for the detection of the disease, it is connected to the monitor through Raspberry-pi with the help of specified USB port. Monitor is used for displaying of captured images as well as output images after detecting the disease and type of the disease.



Fig. 3.11: Schematic of Circuit Set-up

With the help of Wi-Fi module on the Raspberry-pi board, images can be forward to the cloud for storage, to compare the disease affected leaves.

With the help of Image clustering techniques the disease affected part of the image can be extracted and different features will be evaluated. Apart from that the information can also be sent to the concerned client regarding the disease and affected amount so that the amount of pesticides required for the recovery.

3.6 Software Tools

IOT system is nothing but an embedded system which is designed with a combination of hardware and software.

Software tools used in designing the IOT based Leaf Disease Detection are:

- Python IDE
- OpenCV
- MATLAB

Python IDE:

Python IDE is used to write the executable code to get the desired output from the device. Here

By default, Raspbian (Stretch version April 2018 and earlier) uses Python 2. However, versions 2 and 3 come installed by default. We just have to make 1 minor change so that the Pi uses Python 3 whenever we type python into a terminal.

Why only python?

There are other OS's available to the raspberry pi on the ROM site which require or can allow better use for other languages, but the main reason why python is preferred for use on the raspberry pi is because it is a lab-on-a-chip where its more for educational use than anything where we all know python is a fairly easy language to pick up. But in saying that you can find libraries which enable functionality and books on how to use other languages on the raspberry pi like c/c++ for example.

OpenCV:

OpenCV (Open Source Computer Vision) is a library of programming functions mainly aimed at real-time computer vision. In simple language it is library used for Image Processing. It is mainly used to do all the operation related to Images. Since Raspberry Pi's are preloaded with Python, OpenCV can be installed directly.

MATLAB:

MATLAB is a general-purpose programming language. When it is used to process images one can generally write function files, or script files to perform the operations. These files form a formal record of the processing used and ensures that the final results can be tested and replicated by others should the need arise.

In MATLAB, the IPT is a collection of functions that extends the capability of the MATLAB numeric computing environment. It provides a comprehensive set of reference-standard algorithms and workflow applications for image processing, analysis, visualization and algorithm development.

CHAPTER - 4

RESULTS AND DISCUSSIONS

4.1 Process of Execution

For the implementation of the proposed method, number of images are to be collected for reference. Here a reference image is taken as an input image as shown in the figure 4.1.

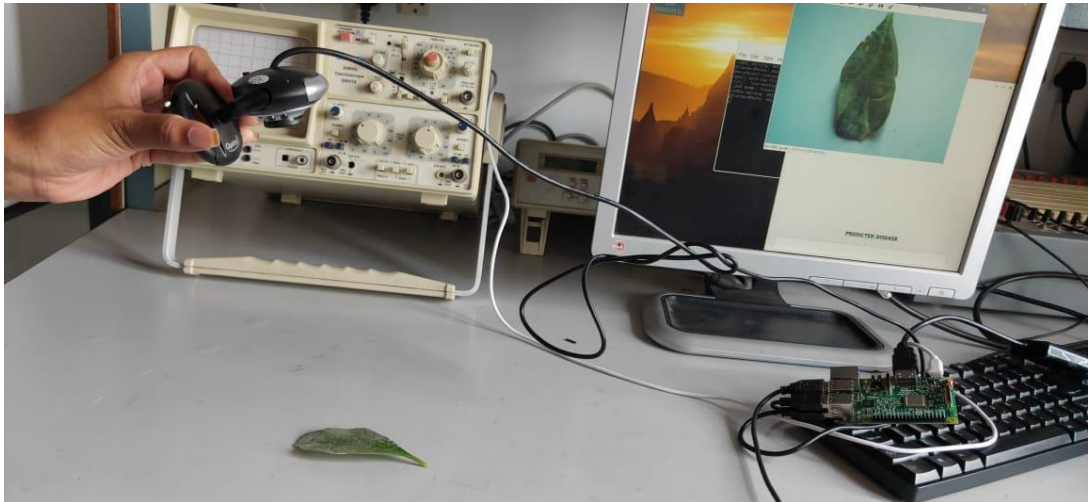


Fig. 4.1: Manual Input Image

An alternatively, the image can also be uploaded from the predefined set unlike captured by camera as shown in fig.4.1 for the analysis of an image. Detection of the disease of the leaf can be identified by comparing the captured image with the pre-defined images (offline database). The process of taking an image from the offline data base is shown in below fig.4.2.

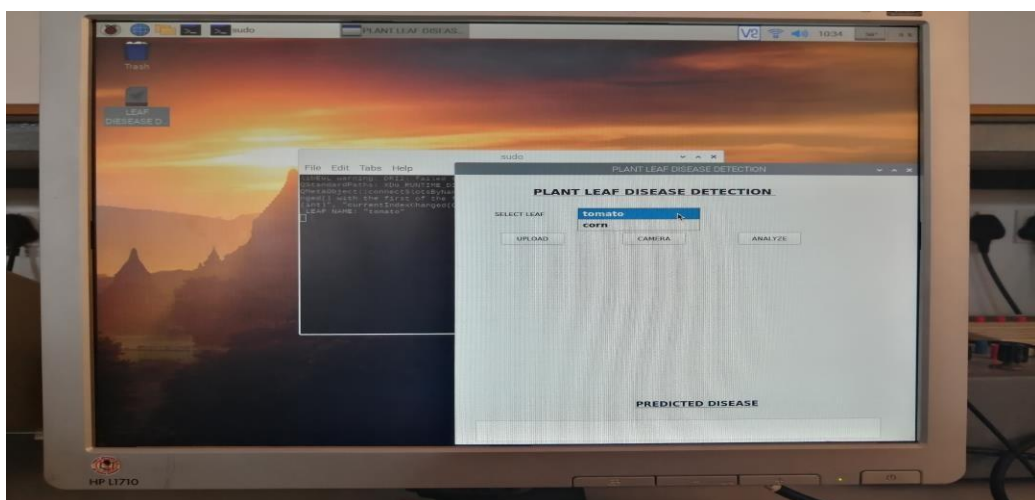


Fig. 4.2: Selecting the input type of plant from the offline database

For the analysis, for an example here a un-healthy corn leaf is taken as input image which is shown below in fig.4.3.

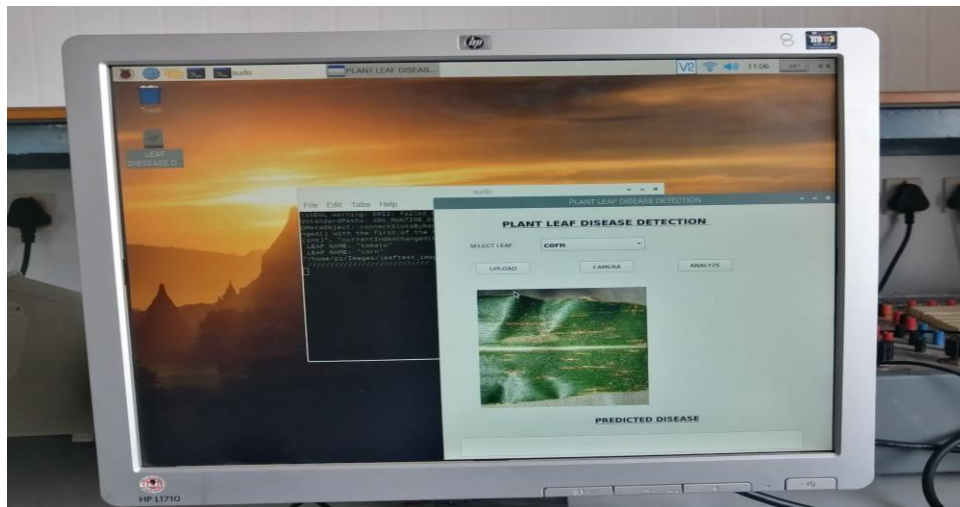


Fig. 4.3: Taking the unhealthy Corn leaf

After uploading a unhealthy corn image, the disease type that occurred to the leaf is predicted by comparing the data from the offline server. And it is observed that it shows the disease type effected to the leaf as shown in fig.4.4. Likely, the disease detection for any type of leaf will be implemented.

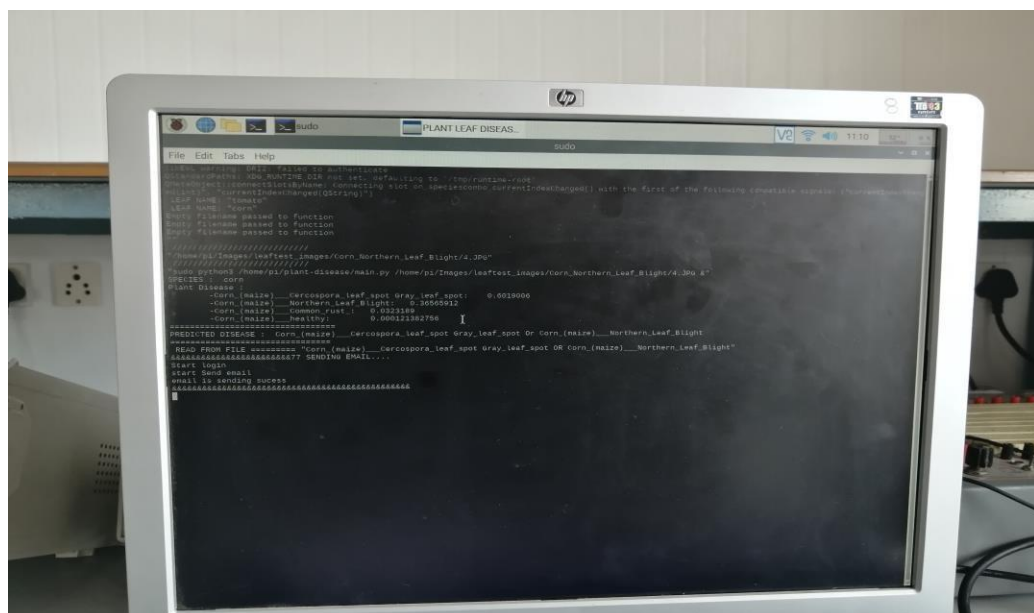


Fig. 4.4: Detection of disease type

4.2 Analysis

However, after the detection of the disease it is to be analyzed that what are the features (area, shape, perimeter, density, etc.) are discriminated from the original image. So, here image analysis will be implemented to estimate how much area is affected with the disease. For that purpose, the image is segmented (clustered) to identify healthy and unhealthy parts of the leaf. So, as stated in the proposal, here the input image is clustered as shown below in the fig.4.5.



Fig. 4.5: Clustering and Area calculation

Based on the clusters generated based on k-means clustering method, it can be observed that identification of leaf affected area is easily predicted when compared with conventional methods with less computation time as shown in fig. 4.5.

CHAPTER - 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

Leaf is the main part of the plant. So, analyzing any disease effective part of leaf is a quite complex task. So, to depict clearer images an efficient imaging technique is chosen. For further analysis of the leaf, an efficient method i.e., k-means clustering algorithm is implemented on captured images for the detection and analysis. Hence it is concluded that the proposed method is computationally efficient when compared with the conventional techniques.

5.2 Future Scope

Designed Leaf Disease Detection device that works on image processing and IOT principles will help in finding out the affected diseases and sends the relevant information to the concerned. This can be extended as a device that is capable of monitoring the field continuously over large areas and that gives data regarding the disease type and the pesticides to be used for that type of disease along with the amount of pesticide. Moreover, the present leading dangerous Viral pneumonia covid-19 threat i.e., corona virus effected images can also be analyzed in an simple manner and also can be estimated effectively how much it is effected with the help of proposed image processing clustering concept. And not only that with the help of IOT environment it is possible to find the people who are affected with Corona like Aarogya Setu.

REFERENCES

REFERENCES

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APPENDIX

Appendix

Main Program:

```
import os

import argparse

import subprocess

import numpy as np

import cv2

import numpy as np

from PIL import Image

from tensorflow.python.keras.models import load_model

from tensorflow.python.keras.preprocessing import image

from tensorflow.python.keras.applications.inception_v3 import preprocess_input

CORN = 'corn'

TOMATO = 'tomato'

SPECIES = [CORN,TOMATO]

DISEASE_SUPPORTED_SPECIES = {CORN,TOMATO}

CORN_CLASSES = ['Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot',
'Corn_(maize)___Common_rust_',
                'Corn_(maize)___Northern_Leaf_Blight', 'Corn_(maize)___healthy']

TOMATO_CLASSES = ['Tomato___Bacterial_spot', 'Tomato___Early_blight',
'Tomato___Late_blight', 'Tomato___Leaf_Mold',
                'Tomato___Septoria_leaf_spot', 'Tomato___Spider_mites Two-
spotted_spider_mite',
```



```

        'Tomato___Target_Spot',
'Tomato___Tomato_Yellow_Leaf_Curl_Virus',
'Tomato___Tomato_mosaic_virus',32

        'Tomato___healthy']

PLANT_CLASSES = {

    CORN: CORN_CLASSES,

    TOMATO: TOMATO_CLASSES

}

VGG_ARCHITECTURE = 'vgg'

INCEPTIONV3_ARCHITECTURE = 'inceptionv3'

SUPPORTED_MODEL_TYPES = {VGG_ARCHITECTURE,
INCEPTIONV3_ARCHITECTURE}

DISEASE_DETECTION = 'disease_detection'

TARGET_IMAGE_SIZES = {

    VGG_ARCHITECTURE: {

        DISEASE_DETECTION: (64, 64),

    },

    INCEPTIONV3_ARCHITECTURE: {

        DISEASE_DETECTION: (100, 100),

    }

}

VGG_MODELS = {

    CORN: '/home/pi/plant-disease/Models/Corn_0.8926_VGG.h5',

```

```

TOMATO: '/home/pi/plant-disease/Models/Tomato_0.8675_VGG.h5'

}

INCEPTIONV3_MODELS = {

    CORN: '/home/pi/plant-disease/Models/InceptionV3-scratch_segCorn.h5',

    TOMATO: '/home/pi/plant-disease/Models/InceptionV3-
scratch_segTomato.h5'

}

MODEL_STORAGE_BASE = '/home/pi/plant-disease/Models'

def get_classes(species_name):

    return PLANT_CLASSES[species_name]

def get_disease_model(species, model_type):

    if species not in DISEASE_SUPPORTED_SPECIES:

        raise ValueError("`{}` species has no disease model yet.\n"

            "Species tha have disease models are {}".format(species,
DISEASE_SUPPORTED_SPECIES))

    if model_type == VGG_ARCHITECTURE:

        return VGG_MODELS[species]

    elif model_type == INCEPTIONV3_ARCHITECTURE:

        return INCEPTIONV3_MODELS[species]

    else:

        raise ValueError("No such `{}` model type is supported.\n"

            "Supported model types are {}".format(model_type,
SUPPORTED_MODEL_TYPES))

```

```

def get_predictions(model_path, img_path, img_target_size):

    if not os.path.exists(model_path):

        raise ValueError('No such `{}` file found\n'

                           'Please, checkout the readme of the project '

                           'on github and download required models'.format(model_path))

    model = load_model(model_path)

    pil_img = Image.open(img_path)

    if pil_img.size != img_target_size:

        pil_img = pil_img.resize(img_target_size)

    img = image.img_to_array(pil_img)

    if img.shape[2] == 4:

        img = img[:, :, :3]

    img = np.expand_dims(img, axis=0)

    img = preprocess_input(img)

    preds = model.predict(img).flatten()

    value_ = preds.argsort()

    sorted_preds_index = value_[::-1]

    return preds, sorted_preds_index

def segment_image(img_path):

    image_name, extension = os.path.splitext(img_path)

    segmented_image_name = image_name + "_marked" + extension # the future
segmented image name to be

```

```

    result = subprocess.check_output(['python', "leaf-image-
segmentation/segment.py", "-s", img_path])

    print('Info: Input image segmented.')

    return segmented_image_name

def predict_disease(img_path, species, model_type=VGG_ARCHITECTURE,
do_print=True):

    if species not in SPECIES:

        raise ValueError("No such `{}` species is supported.\n"

                        "Supported species are {}".format(species, SPECIES))

    if species not in DISEASE_SUPPORTED_SPECIES:

        print("Info: For `{}` species, a disease can not be predicted "

              "since its disease model is not implemented yet.".format(species))

        return None

    else:

        SPECIES_CLASSES = get_classes(species)

        model_path = os.path.join(MODEL_STORAGE_BASE,
get_disease_model(species, model_type))

        target_image_size =
TARGET_IMAGE_SIZES[model_type][DISEASE_DETECTION]

        preds, sorted_preds_index = get_predictions(model_path, img_path,
target_image_size)

        if do_print:

            print("Plant Disease : ")

            for i in sorted_preds_index:

```

```

        print("\t-" + str(SPECIES_CLASSES[i]) + ": \t" + str(preds[i]))

    return
str(SPECIES_CLASSES[sorted_preds_index[0]]),str(SPECIES_CLASSES[sorted_preds_index[1]])

def get_cmd_args():

    parser = argparse.ArgumentParser()

    parser.add_argument("image", type=str, help='Image file path')

    args = parser.parse_args()

    return args

if __name__ == "__main__":

    args = get_cmd_args()

    f = open("/home/pi/build-leafdiseasegui/species.txt", "r")

    species=f.readline()

    print("SPECIES : ", species)

    f.close()

    diseasereturn1,diseasereturn2=predict_disease(args.image, species)

    print("=====")

    print("PREDICTED DISEASE : ", diseasereturn1 , "Or", diseasereturn2)

    print("=====")

    f = open("/home/pi/build-leafdiseasegui/diseasename.txt", "w")

    f.write(diseasereturn1)

    f.write(" OR ")

    f.write(diseasereturn2)

```

```

f.close()

f = open("/home/pi/build-leafdiseasegui/readfilebit.txt", "w")

f.write("1")

f.close()

```

Code for finding Effectuated Area Percentage:

```
% Project Title: Pomegranate Leaf Disease Detection
```

```
clc
```

```
close all
```

```
clear all
```

```
[filename, pathname] = uigetfile({'*.*'; '*.bmp'; '*.jpg'; '*.gif'}, 'Pick a Leaf Image File');
```

```
I = imread([pathname,filename]);
```

```
I = imresize(I,[256,256]);
```

```
%figure, imshow(I); title('Query Leaf Image');
```

```
% Enhance Contrast
```

```
I = imadjust(I,stretchlim(I));
```

```
figure, imshow(I);title('Contrast Enhanced');
```

```
% Otsu Segmentation
```

```
I_Otsu = im2bw(I,graythresh(I));
```

```
% Conversion to HIS
```

```
I_HIS = rgb2hsi(I);
```

```
%% Extract Features
```

```

% Function call to evaluate features
%[feat_disease seg_img] = EvaluateFeatures(I)

% Color Image Segmentation
% Use of K Means clustering for segmentation
% Convert Image from RGB Color Space to L*a*b* Color Space
% The L*a*b* space consists of a luminosity layer 'L*', chromaticity-layer 'a*'
and 'b*'.
% All of the color information is in the 'a*' and 'b*' layers.
cform = makecform('srgb2lab');
% Apply the colorform
lab_he = applycform(I,cform);

% Classify the colors in a*b* colorspace using K means clustering.
% Since the image has 3 colors create 3 clusters.
% Measure the distance using Euclidean Distance Metric.
ab = double(lab_he(:,2:3));
nrows = size(ab,1);
ncols = size(ab,2);
ab = reshape(ab,nrows*ncols,2);
nColors = 3;
[cluster_idx cluster_center] = kmeans(ab,nColors,'distance','sqEuclidean', ...
                                     'Replicates',3);
%[cluster_idx cluster_center] =
kmeans(ab,nColors,'distance','sqEuclidean','Replicates',3);
% Label every pixel in the image using results from K means
pixel_labels = reshape(cluster_idx,nrows,ncols);
%figure,imshow(pixel_labels,[]), title('Image Labeled by Cluster Index');

% Create a blank cell array to store the results of clustering
segmented_images = cell(1,3);
% Create RGB label using pixel_labels
rgb_label = repmat(pixel_labels,[1,1,3]);

```

```

for k = 1:nColors
    colors = I;
    colors(rgb_label ~= k) = 0;
    segmented_images{k} = colors;
end
figure, subplot(3,1,1);imshow(segmented_images{1});title('Cluster 1');
subplot(3,1,2);imshow(segmented_images{2});title('Cluster 2');
subplot(3,1,3);imshow(segmented_images{3});title('Cluster 3');
set(gcf, 'Position', get(0,'Screensize'));

% Feature Extraction
x = inputdlg('Enter the cluster no. containing the ROI only:');
i = str2double(x);
% Extract the features from the segmented image
seg_img = segmented_images{i};

% Convert to grayscale if image is RGB
if ndims(seg_img) == 3
    img = rgb2gray(seg_img);
end
%figure, imshow(img); title('Gray Scale Image');

% Evaluate the disease affected area
black = im2bw(seg_img,graythresh(seg_img));
%figure, imshow(black);title('Black & White Image');
m = size(seg_img,1);
n = size(seg_img,2);

zero_image = zeros(m,n);
%G = imoverlay(zero_image,seg_img,[1 0 0]);

cc = bwconncomp(seg_img,6);

```



```

diseasedata = regionprops(cc,'basic');
A1 = diseasedata.Area;
sprintf('Area of the disease affected region is : %g%',A1);

I_black = im2bw(I,graythresh(I));
kk = bwconncomp(I,6);
leafdata = regionprops(kk,'basic');
A2 = leafdata.Area;
sprintf(' Total leaf area is : %g%',A2);

%Affected_Area = 1-(A1/A2);
Affected_Area = (A1/A2);
if Affected_Area < 0.1
    Affected_Area = Affected_Area+0.15;
end
sprintf('Affected Area is: %g%%',(Affected_Area*100))

% Create the Gray Level Cooccurrence Matrices (GLCMs)
glcms = graycomatrix(img);

% Derive Statistics from GLCM
stats = graycoprops(glcms,'Contrast Correlation Energy Homogeneity');
Contrast = stats.Contrast;
Correlation = stats.Correlation;
Energy = stats.Energy;
Homogeneity = stats.Homogeneity;
Mean = mean2(seg_img);
Standard_Deviation = std2(seg_img);
Entropy = entropy(seg_img);
RMS = mean2(rms(seg_img));
%Skewness = skewness(img)
Variance = mean2(var(double(seg_img)));

```

```

a = sum(double(seg_img(:)));
Smoothness = 1-(1/(1+a));
Kurtosis = kurtosis(double(seg_img(:)));
Skewness = skewness(double(seg_img(:)));
% Inverse Difference Movement
m = size(seg_img,1);
n = size(seg_img,2);
in_diff = 0;
for i = 1:m
    for j = 1:n
        temp = seg_img(i,j)./(1+(i-j).^2);
        in_diff = in_diff+temp;
    end
end
IDM = double(in_diff);

feat_disease = [Contrast,Correlation,Energy,Homogeneity, Mean,
Standard_Deviation, Entropy, RMS, Variance, Smoothness, Kurtosis, Skewness,
IDM];
%%
% Load All The Features
load('Training_Data.mat')

% Put the test features into variable 'test'
test = feat_disease;
result = multisvm(Train_Feat,Train_Label,test);
%disp(result);

% Visualize Results
if result == 0
    helpdlg(' Alternaria Alternata ');
    disp(' Alternaria Alternata ');
elseif result == 1

```

```

        helpdlg(' Anthracnose ');
        disp('Anthracnose');
elseif result == 2
    helpdlg(' Bacterial Blight ');
    disp(' Bacterial Blight ');
elseif result == 3
    helpdlg(' Cercospora Leaf Spot ');
    disp('Cercospora Leaf Spot');
elseif result == 4
    helpdlg(' Healthy Leaf ');
    disp('Healthy Leaf ');
end

%% Evaluate Accuracy
load('Accuracy_Data.mat')
Accuracy_Percent= zeros(200,1);
for i = 1:500
    data = Train_Feat;
    %groups = ismember(Train_Label,1);
    groups = ismember(Train_Label,0);
    [train,test] = crossvalind('HoldOut',groups);
    cp = classperf(groups);
    svmStruct =
    svmtrain(data(train,:),groups(train),'showplot',false,'kernel_function','linear');
    classes = svmclassify(svmStruct,data(test,:), 'showplot',false);
    classperf(cp,classes,test);
    Accuracy = cp.CorrectRate;
    Accuracy_Percent(i) = Accuracy.*100;
end
Max_Accuracy = max(Accuracy_Percent);
sprintf('Accuracy of Linear Kernel with 500 iterations is:
%g%%',Max_Accuracy)

```

Camera Code:

```
import cv2

import os

cam = cv2.VideoCapture(0)

cv2.namedWindow("test")

img_counter = 0

savepath="/home/pi/Images"

while True:

    ret, frame = cam.read()

    cv2.imshow("test", frame)

    if not ret:

        break

    k = cv2.waitKey(1)

    if k%256 == 27:

        # ESC pressed

        print("Escape hit, closing...")

        break

    elif k%256 == 32:

        # SPACE pressed

        img_name = "leaf_cameraimage_{}.jpg".format(img_counter)

        cv2.imwrite(os.path.join(savepath,img_name), frame)
```

```
print("{} written!".format(img_name))  
  
img_counter += 1  
  
cam.release()  
  
cv2.destroyAllWindows()
```